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# Saliency Prediction with Conditional GAN

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**TED UNIVERSITY**

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# 1 Introduction

Visually salient stimuli draw our attention. In a cluttered visual world, complex biological systems need to be able to quickly detect potential prey, predators, or mates. However, even for the most advanced biological brains, simultaneously identifying all interesting targets in one's visual field has prohibitive computational complexity, making it a difficult task. Visual attention could be a way to compensate for the inability to fully process all locations at the same time [18]. This solution, however, brings a challenge. How do you choose the focus of your attention if you only have time to process one region or object at a time? Visual salience aids your brain in making reasonable selections. The spatial locations in an image that attract human attention are referred to as visually salient regions.

The visual attention mechanism is heavily focused on visual saliency. Bottom-up focus and top-down attention are the two primary forms of visual attention. Top-down attention is goal-oriented and focused on what is perceived, while bottom-up attention is guided by the properties of visual scenes regardless of substance. When individuals are free to perceive sensory objects, bottom-up perception occurs. Low-level features (such as color contrast, depth, motion, object centeredness, and so on) are critical for the visual focus system in this situation. When people see visual feedback relevant to the content, such as "important person in the picture", "what behavior the people in the image are doing", and so on, top-down focus occurs. High-level features (such as whether the frame includes individuals or not, what feelings are present in the frame, etc.) are essential for the visual focus system if the watching act has a content related objective.

The visual saliency methods produce

saliency maps that estimate the likelihood of being fixated at each pixel. Visual saliency maps are predicted by algorithms that can use hand-crafted visual stimulus features or some features can be trained through training of learning methods, and visual saliency estimation is achieved using the learned parameters.

In this work our goal is to explore beyond the [1]'s work, to propose a single method for accurately predicting saliency maps for both bottom-up and top-down perception. Inspired by our cognitive system we thought that if the machine was able to segmentate an image and make a correlation with some segments and saliency maps that are taken as ground truth, it would be more successful for predicting the saliency maps for the tasks that were in [1]. Since the aforementioned tasks are about humans, like how many are there in the image or about their expressions, we will use only humans as a new channel after RGB channels. We will train our model for segmented images, that were segmented by a pre-trained model that is known to be successful overall. After that we will use segmented humans as a new binary channel on top of RGB images. Another layer of our work will be adding faces so that our input to our Conditional Generative Adversarial Network (CGAN) structure RGBP images where P is a layer that can have three values. Those values will be 0 for unknown, 1 for humans' bodies and 2 for their faces. Of course bodies and faces will be found via pre-trained models since they are not our primary topic. Since we use pre-trained models there may be, and most likely there will be, some errors in human detection and face detection. But as previously mentioned since these are not our primary concerns, we will ignore those errors for now.

## 2 Related Work

The field of saliency prediction has been extensively researched since the first attempts to model human saliency by feature integration. Koch and Ullman [11] and Itti et al. [10] suggested methods focused on extracting image features such as color, intensity, and orientation, which built on this bottom-up approach. Writers in [5, 14] on the other hand, have taken a top-down view, based on information theory, in which task-dependent functions are used. They include features from human detectors and HoG descriptors, as well as local information from regions of interest for the task at hand. These models [5, 14] are entirely task-specific, making it almost impossible to adapt from one task to the next.

Hand-crafted features became less desirable as deep learning methods for visual saliency solutions became more popular. Several attempts [7, 6] to model the visual saliency of the human free viewing system with deep convolutional networks have been made in recent years, inspired by the success of deep learning techniques. Convolutional Neural Networks have shown to be effective in a variety of computer vision areas, including visual saliency. CNN methods have the advantage of learning and/or introducing important features rather than utilizing previously defined hand-crafted features, resulting in more detailed saliency charts. As an example, DeepGaze [13] is a well-known CNN structure for visual saliency that employs five layers. Unlike the majority of deep learning solutions, Deepfix [12] uses Fully Connected Neural Networks (FCNN), which is a more recently proposed popular solution. CNN, on the other hand, necessarily involves a learning process, as well as data to train and processing resources, which can be seen as a drawback.

As an alternative Goodfellow et al. [8] introduced a specific unsupervised deep

learning approach called Generative Adversarial Nets (GAN). Generative Adversarial Network (GAN)s, which are part of the generative model family, have shown promise in pix-to-pix synthesis [2]. When synthesizing natural pictures, some studies have looked at a variety of architectural augmentations. Authors in [16] used the GAN architecture to forecast visual saliency and suggested a novel loss feature that has been shown to be useful for both initializing the generator and stabilizing adversarial preparation. Rather than developing a customized loss feature, [12] propose SalGAN, an adversarial training method for visual saliency prediction motivated by generative adversarial networks (GANs). SalGAN uses a deep convolutional neural network to estimate the saliency map of an input image (DCNN). This network is first trained on down-sampled versions of the saliency maps using a binary cross-entropy (BCE) loss. The model is then fine-tuned using a discriminator network trained to distinguish between the saliency maps created by SalGAN and the actual ones used as ground truth in a binary classification task. Also [1] had proposed to use a single CGAN architecture for both bottom-up and top-down visual saliency prediction tasks.

If both the generator and the discriminator are conditioned on some additional information  $y$ , such as class labels or input from other modalities, generative adversarial networks can be expanded to a conditional model. Conditioning can be done by putting  $y$  into the discriminator and generator as an additional input layer. The previous input noise  $p_z(z)$  and  $y$  are mixed in a joint hidden representation in the generator, and the adversarial training framework provides for a lot of freedom in how this hidden representation is built.[15] You can see a basic CGAN structure in Figure 1.

### 3 Proposed Method

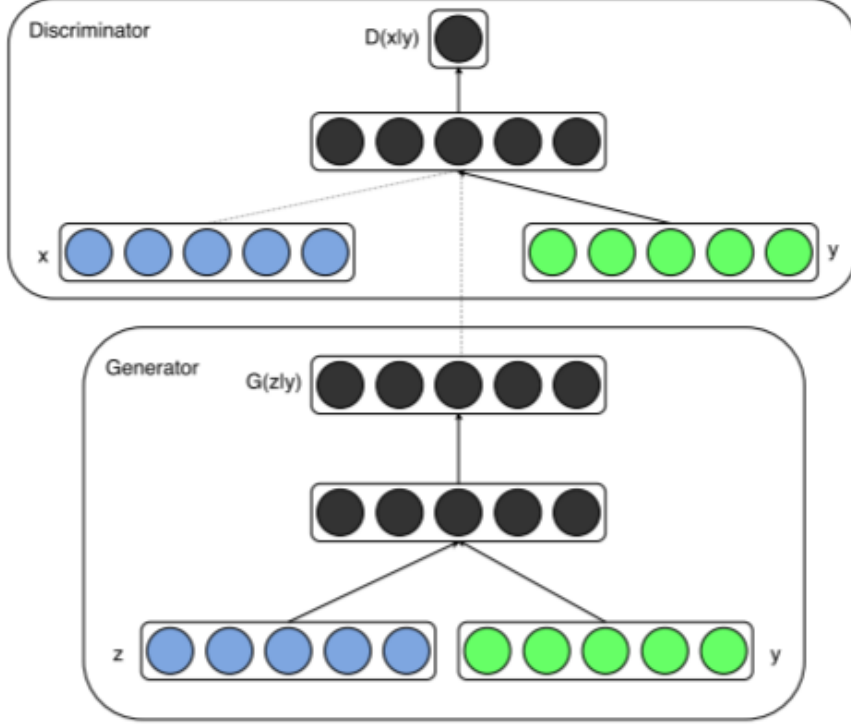


Figure 1: Conditional Adversarial Net[15]

GANs are generative models that learn a mapping from random noise vector  $z$  to output image  $y$ ,  $G : z \rightarrow y$ [8]. **CGANs**, on the other hand, train a mapping from the observed picture  $x$  and the random noise vector  $z$  to  $y$ ,  $G: x, z, y$ . An adversarially trained discriminator,  $D$ , is trained to do as good as possible at recognizing the generator’s "fakes".[9] The generator  $G$  is trained to produce outputs that cannot be distinguished from "genuine" images. We will use stimuli images, human labeled images and outputs of face detection models. The objective of **GANs** can be represented as :

$$\min_{\theta} \max_{\phi} V(G_{\theta}, D_{\phi}) = \mathbb{E}_{\mathbf{x} \sim \mathbf{p}_{\text{data}}} [\log D_{\phi}(\mathbf{x})] + \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} [\log(1 - D_{\phi}(G_{\theta}(\mathbf{z})))] \quad (1)$$

### 3.1 Network Architecture

We modify the architectures of our generator and discriminator based on [9, 17]. Convolution2-BatchNorm-ReLu mod-

ules are used by both the generator and discriminator. One of the most prevalent architectures for this type of problem is encoder-decoder networks. The generator is based on a U-Net design and is an encoder-decoder model. The model starts with a source picture (such as an RGB picture) and creates a target picture from it (e.g. Eye-Fixation Map). It accomplishes this by downsampling or encoding the input picture to a bottleneck layer, then upsampling or decoding the bottleneck representation to the output picture size. The input is transmitted via a number of downsampling layers until it reaches a bottleneck layer, where the process is reversed. The U-Net design adds skip-connections between the encoding and decoding layers, producing a U-shape. All information must pass through all layers of the network, including the bottleneck, in order for it to function. There is a lot of low-level information shared between the input and output for many picture translation difficulties, and it would be preferable to send it directly via the network.

## 4 Experiments

### 4.1 Dataset

The dataset we have used in this experiment is the same as the dataset used in [1]. The subjects' gaze data was collected using a SMI eye tracker model SMI RED-60 Hz with a monitor linked to a DELL laptop. The participants looked at still images on a 22-inch monitor, and an eye tracker at

the bottom of the monitor recorded their fixation data as they watched 2. Each task-based watching and free-viewing session had eight individuals; in total, 32 people took part in our study. We used 32 participants with a mean age of  $25,13 \pm 3,86$ , 13 of whom are female and 19 of whom are male, all of whom had normal or corrected-to-normal visual acuity [1].



Figure 2: Viewing monitor with SMI eye tracker[1].  
<https://imotions.com/hardware/smi-red/>

There are 492 stimuli images that are collected. There were 4 tasks for the viewers:

- For task 1, they asked the participants to count as many people as they could in the frame, if any, and come up with a number, which did not have to be accurate, that represented the quantity of persons they could count.[1]
- They asked the participants to answer questions about emotion in the frame and people's feelings in the frame for task 2. Before the experiment, students were given some examples such as "sad," "happy," "angry," and so on, and they were also free to choose any words for the feelings that came to mind while watching.[1]
- Participants were asked to define the occasion in the frame, if there was one, or what the person or individuals in the frame were doing in task 3.[1]
- Task 4 is free-viewing, in which the participants were not given any instructions and were only told to look at the visuals provided.[1]

### 4.2 Pre-Trained Models

As we have previously mentioned there are two ready to use models that we have used in this research. The model we have used for image segmentation is-

DeeplabV3+ResNeSt Semantic Segmentation Model [4, 19] which is trained with ADE20K dataset [20]. It is retrieved from [Gluon's Model Zoo](#). So, we will have the output of this model and save it so that every part labeled as "human/person" will be

white and all others as black. We are only using "human/person" label since our tasks are mostly about humans in the images.

Another pre-trained model that we have used is for detecting faces in the images. It is [DLIB's Frontal Face Detector](#). Similar to the previous one we will be saving "detected" faces as white and other parts as black.

As we have mentioned before we are using these pre-trained models because their functionalities are not our main focus in this research. We are more interested in their effects on our [CGAN's](#) outputs' accuracy. So, even though we know that these models are not perfect or very accurate, we believe that their accuracy will be enough for this research. In this research we aim to make a new way to new researches. Other reason we have used these models are, this research is for an under-

graduate course, so we have a time limit and searching for or training close to perfect models for these tasks will consume most of our times.

### 4.3 Evaluation Methods

We will be using similarity metrics for comparing generated eye-fixation maps with the ones that we accept as ground truths which were acquired during [\[1\]](#)'s work. The metrics we will use are Area Under the Curve ([AUC](#))-Borji [\[3\]](#), Similarity (SIM)-histogram intersection-, Pearson's linear coefficient (CC) and normalized scanpath saliency (Normalized Scanpath Saliency ([NSS](#))). High the results that we obtain from these metrics mean that using our proposed method, using a new channel which contains "potential" persons and "potential" their faces, improves results for any task while predicting saliency maps.

## 5 Results

We will display our results in this sections. From now on we will use RGB for the the results obtained from the model that we have only used only the original RGB image that is in our dataset. RGBP means that we have added another layer to our RGB images and used that 4 channeled image as input for the training and testing of our model. This layer has "1"s as value where our model detected human/person and "0"s in all other pixels. RGBPF means that we have used again a 4-channeled image but in this format 4<sup>th</sup> channel has "2"s as values of pixels where our pre-trained face detection model detected faces. So, in RGBPF, "2"s represent faces, "1"s for other body parts of persons'/humans' and "0" for other. We will use aforementioned abbreviations along with tasks. For example "Task 1 - RGBPF" means that the result is obtained from a model that is trained with Fixation Density Map ([FDM](#)) of images' in Task 1 and RGBPF of the corresponding image that has 4 channel and has faces and humans marked in the last channel.

What we are trying to achieve is a new, promising method that can be used for similar tasks. We are aiming to observe improvements in the scores as we add the new layers, Humans and later on top of that faces, to our plain RGB images. So, if we achieve consisten improvements in the scores as we add humans and faces into the new channel, I believe we can say we have achieved our goal considering the limited time and resources we had. For all the tables best results in each column will be **bold**.



## 5.1 Results for Task 1

In this task participants were asked to count the number of people in the image. For this task we expect to see that results will improve especially after we add humans layer. We are doubtful that adding faces for this task will improve results significantly.

|                | <b>AUC Borji</b> | <b>KL</b>       | <b>NSS</b>      | <b>SIM</b>      | <b>CC</b>       |
|----------------|------------------|-----------------|-----------------|-----------------|-----------------|
| Task 1 - RGB   | 0.519634         | 8.726078        | 0.801986        | 0.370632        | 0.38747         |
| Task 1 - RGBP  | <b>0.527001</b>  | 7.105701        | 0.897313        | 0.420876        | 0.464952        |
| Task 1 - RGBPF | 0.524533         | <b>6.540584</b> | <b>0.952212</b> | <b>0.439577</b> | <b>0.486119</b> |

Table 1: Average results for model trained for Task 1

In the following table you can see the images that has the best results for each metric.

|                | <b>AUC Borji</b> | <b>KL</b> | <b>NSS</b> | <b>SIM</b> | <b>CC</b> |
|----------------|------------------|-----------|------------|------------|-----------|
| Task 1 - RGB   | 0.55925          | 0.30449   | 1.5745     | 0.92143    | 0.98682   |
| Task 1 - RGBP  | 0.59678          | 0.51065   | 2.0365     | 0.89976    | 0.97897   |
| Task 1 - RGBPF | 0.58647          | 0.99021   | 1.8504     | 0.88851    | 0.9743    |

Table 2: Best results for each metric for Task 1.

You can find the worst results that we got for the Task 1 in the table below.

|                | <b>AUC Borji</b> | <b>KL</b> | <b>NSS</b> | <b>SIM</b> | <b>CC</b> |
|----------------|------------------|-----------|------------|------------|-----------|
| Task 1 - RGB   | 0.49782          | 25.2023   | -0.03691   | 0.047746   | -0.02702  |
| Task 1 - RGBP  | 0.50371          | 25.4516   | 0.0815     | 0.069766   | 0.01      |
| Task 1 - RGBPF | 0.50184          | 24.1932   | 0.27055    | 0.10575    | 0.057787  |

Table 3: Worst results for each metric for Task 1.

## 5.2 Results for Task 2

For this task people were asked to answer questions about the emotions in the image. We are expecting to see better results for each addition we make, i.e. we expect to see better results for RGBP than RGB and RGBPF should have the best results among these.

|                | <b>AUC Borji</b> | <b>KL</b>       | <b>NSS</b>      | <b>SIM</b>      | <b>CC</b>       |
|----------------|------------------|-----------------|-----------------|-----------------|-----------------|
| Task 2 - RGB   | 0.517409         | 8.124514        | 0.820892        | 0.359436        | 0.391114        |
| Task 2 - RGBP  | 0.520002         | <b>7.769992</b> | 0.875991        | 0.40646         | 0.468935        |
| Task 2 - RGBPF | <b>0.520976</b>  | 8.066703        | <b>0.895648</b> | <b>0.407357</b> | <b>0.472414</b> |

Table 4: Results table for the model trained for Task 2

Following table shows us the name of the images that has the best results.

|                | <b>AUC Borji</b> | <b>KL</b> | <b>NSS</b> | <b>SIM</b> | <b>CC</b> |
|----------------|------------------|-----------|------------|------------|-----------|
| Task 2 - RGB   | 0.55126          | 0.27777   | 1.8601     | 0.89961    | 0.97953   |
| Task 2 - RGBP  | 0.55412          | 1.4118    | 1.8406     | 0.84431    | 0.97329   |
| Task 2 - RGBPF | 0.57677          | 0.69642   | 2.1048     | 0.86059    | 0.97202   |

Table 5: Best results for each metric for Task 2.

In the following table you can find the worst results we get.

|                | <b>AUC Borji</b> | <b>KL</b> | <b>NSS</b> | <b>SIM</b> | <b>CC</b> |
|----------------|------------------|-----------|------------|------------|-----------|
| Task 2 - RGB   | 0.49764          | 22.9733   | 0.18428    | 0.10145    | 0.017562  |
| Task 2 - RGBP  | 0.49898          | 22.0774   | 0.17107    | 0.1318     | 0.036406  |
| Task 2 - RGBPF | 0.50081          | 22.5294   | 0.23401    | 0.15288    | 0.11029   |

Table 6: Worst results for each metric for Task 2.

### 5.3 Results for Task 3

In this task participants were asked to define the occasion in the frame. Again we are expecting to see that results will improve especially after we add humans layer.

|                | <b>AUC Borji</b> | <b>KL</b>       | <b>NSS</b>      | <b>SIM</b>      | <b>CC</b>       |
|----------------|------------------|-----------------|-----------------|-----------------|-----------------|
| Task 2 - RGB   | 0.517409         | 8.124514        | 0.820892        | 0.359436        | 0.391114        |
| Task 2 - RGBP  | 0.520002         | <b>7.769992</b> | 0.875991        | 0.40646         | 0.468935        |
| Task 2 - RGBPF | <b>0.520976</b>  | 8.066703        | <b>0.895648</b> | <b>0.407357</b> | <b>0.472414</b> |

Table 7: Results table for the model trained for Task 3

In the table below you can see the best results we got for each of the metrics.

|                | <b>AUC Borji</b> | <b>KL</b> | <b>NSS</b> | <b>SIM</b> | <b>CC</b> |
|----------------|------------------|-----------|------------|------------|-----------|
| Task 3 - RGB   | 0.59392          | 0.51712   | 2.3324     | 0.88288    | 0.97806   |
| Task 3 - RGBP  | 0.54827          | 0.96827   | 1.5675     | 0.8453     | 0.96123   |
| Task 3 - RGBPF | 0.56726          | 1.3529    | 2.1843     | 0.78214    | 0.95317   |

Table 8: Best results for each metric for Task 3.

You can find the worst results we have acquired in the following table.

|                | <b>AUC Borji</b> | <b>KL</b> | <b>NSS</b> | <b>SIM</b> | <b>CC</b> |
|----------------|------------------|-----------|------------|------------|-----------|
| Task 3 - RGB   | 0.50271          | 24.5512   | 0.082419   | 0.075956   | 0.00089   |
| Task 3 - RGBP  | 0.50185          | 17.0679   | 0.26001    | 0.18331    | 0.10567   |
| Task 3 - RGBPF | 0.50415          | 24.0274   | 0.30152    | 0.12237    | 0.056819  |

Table 9: Worst results for each metric for Task 3.

## 5.4 Results for Task 4

In this task participants were asked to view image freely. We believe that while viewing image freely, if there is one or more people in the image people will observe them. We expect to see some improvements in the results as we add layers, but we are doubtful that the change in the results would be significant as the other tasks.

|                | <b>AUC Borji</b> | <b>KL</b>      | <b>NSS</b>      | <b>SIM</b>      | <b>CC</b>       |
|----------------|------------------|----------------|-----------------|-----------------|-----------------|
| Task 4 - RGB   | 0.512942         | 9.391118       | 0.683166        | 0.348988        | 0.379647        |
| Task 4 - RGBP  | 0.512777         | <b>8.67594</b> | <b>0.734171</b> | <b>0.391323</b> | <b>0.448881</b> |
| Task 4 - RGBPF | <b>0.51434</b>   | 9.246382       | 0.712431        | 0.380088        | 0.436837        |

Table 10: Results table for the model trained for free viewing

In the following table you can find the best results for Task 4.

|                | <b>AUC Borji</b> | <b>KL</b> | <b>NSS</b> | <b>SIM</b> | <b>CC</b> |
|----------------|------------------|-----------|------------|------------|-----------|
| Task 4 - RGB   | 0.5648           | 0.81227   | 1.7986     | 0.89604    | 0.97444   |
| Task 4 - RGBP  | 0.54355          | 0.35519   | 1.4603     | 0.91301    | 0.98255   |
| Task 4 - RGBPF | 0.5456           | 0.36267   | 1.287      | 0.86402    | 0.95983   |

Table 11: Best results for each metric for Task 4.

You can find the worst results for Task 4 in the table below.

|                | <b>AUC Borji</b> | <b>KL</b> | <b>NSS</b> | <b>SIM</b> | <b>CC</b> |
|----------------|------------------|-----------|------------|------------|-----------|
| Task 4 - RGB   | 0.4995           | 20.1047   | 0.21913    | 0.14135    | 0.032183  |
| Task 4 - RGBP  | 0.49905          | 19.5786   | 0.26078    | 0.18287    | 0.12984   |
| Task 4 - RGBPF | 0.50234          | 18.9323   | 0.23145    | 0.17224    | 0.12641   |

Table 12: Worst results for each metric for Task 4.

## 6 Discussion

Our main goal in this research was proposing a new methodology for improving the results obtained for the visual saliency prediction models. In general, I believe that the idea we propose showed a promising result by improving the results with respect to the results obtained by regular images. I think if we have used a **CGAN** model that had been proven to be good for saliency prediction and modified it so that it would be trained with a new channel like the one we have proposed in this work, it would increase its results.

When we compare our results with the other research in this field we can see that we can see that our model is not quite successful. There can be many reasons for this, but I believe the main reason is that we selected to focus on other improvements, then we thought that if we had enough time we can come back and tune the model afterwards. Unfortunately, we did not have enough time to go back and work on our **CGAN** structure further.

Also when, we compare our best and worst results we can see that gap is too wide between them. I believe that this is due to wrong partition of test and train data. We have less data in our data set than other similar works. So, if we could increase the training data and tune the training variables better we could achieve a better and more uniformly distributed results.

## List of Abbreviations

**FDM** Fixation Density Map

**CGAN** Conditional Generative Adversarial Network

**GAN** Generative Adversarial Network

**NSS** Normalized Scanpath Saliency

**AUC** Area Under the Curve

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